Homework 6 - Biostatistics

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June 23rd, 2025

Homework 6

In this assignment, we will compare means between paired and independent groups using t-tests to asses changes in immune cell percentages over time and differences based on IL-2 exposure. We will also analyze cholesterol and bone density data from clinical studies to evaluate the effects of dietary and maternal behaviors on health outcomes.

Q1:

Last time, we tested for a difference between 4 and 72 hour CD8 data by creating a new variable that summarized the difference for each person. You can do the same test using the "paired=T" choice, without creating this variable. Compare the 4 hour and 72 hour CD8 using a paired design.

```
# load the data
blood <- read.csv("Blood over time.csv")

# Extract CD8 measurements at 4hrs and 72hrs
cd8_4h <- subset(blood, Hours.post.draw == 4)[, c("ID", "CD8.xCD127")]
cd8_72h <- subset(blood, Hours.post.draw == 72)[, c("ID", "CD8.xCD127")]

# Rename our columns for clairty
colnames(cd8_4h)[2] <- "CD8_4h"
colnames(cd8_72h)[2] <- "CD8_72h"

# Merge the two datasets by subject ID to pair the measurements
paired_data <- merge(cd8_4h, cd8_72h, by = "ID")

# paired t-test
t.test(paired_data$CD8_4h, paired_data$CD8_72h, paired = "TRUE")
```

```
    Paired t-test

data:  paired_data$CD8_4h and paired_data$CD8_72h
t = 4.1543, df = 39, p-value = 0.0001722
```

```
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 1.877406 5.440393
sample estimates:
mean difference
         3.6589
```

In conclusion, we found a statistically significant decrease in the percentage of CD8+ T cells expressing CD127 between 4 and 72 hours after each blood collection (paired t-test, p=0.00017). On average, CD8+CD127+ expression dropped by 3.66% percentage points over this time period, with a 95% confidence interval ranging from 1.93% to 5.39%. These results support concerns that prolonged storage negatively affects CD8 marker expression, potentially impacting downstream experimental reliability.

Q2:

Some of the individuals in the sample had taken IL-2 in the recent past when their blood was drawn; others had not. Do a t-test to answer the question as to whether there is a difference in the CD4% between people who have and have not taken IL-2, based on the 4 hour data. Write a one-sentence conclusion (including the p-value) of the results of the test, the way you might include it in a science paper. Also give the 95% confidence interval for the difference in means. Do this:

```r
# Load data and do pre-processing
blood2 <- read.csv("Blood over time.csv")

# Filter for 4-hour measurements only
cd4_4h <- subset(blood2, Hours.post.draw ==4)

# Group data by IL-2 status
yes_group <- subset(cd4_4h, IL2 == "YES")$CD4.xCD127
no_group <- subset(cd4_4h, IL2 == "NO")$CD4.xCD127
```

a) Assuming equal variances.

```r
# Unpaired t-test assuming equal variances
t.test(yes_group, no_group, var.equal = TRUE)
```

```
    Two Sample t-test

data:  yes_group and no_group
t = 1.1878, df = 38, p-value = 0.2423
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -2.732568 10.491456
sample estimates:
mean of x mean of y
 84.16330  80.28386
```

There was no statistically significant difference in CD4+CD127+ expression at 4 hours between individuals with and without recent IL-2 exposure (two-sample t-test, equal variances assumed p = 0.2423, alpha = 0.05)

b) Assuming unequal variances.

```
# Unpaired t-test assuming unequal variances
t.test(yes_group, no_group, var.equal = FALSE)
```

```
    Welch Two Sample t-test

data:  yes_group and no_group
t = 1.3494, df = 32.962, p-value = 0.1864
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.970070  9.728957
sample estimates:
mean of x mean of y
 84.16330  80.28386
```

There was no statistically significant difference in CD4+CD127+ expression at 4 hours between individuals with and without recent IL-2 exposure (Welch's t-test, p=0.1864, alpha = 0.05).

c) Minimum Sample Size threshold for a given Statistical power (optional).

The researcher comes back to you a few weeks later and wants to design two new trials to address the same questions, with higher statistical power. How many subjects to answer the following question: Are there differences between people with/without a history of IL2? Assume that CD4% in both groups will be normally distributed with a standard deviation of 8%. How many people are needed PER GROUP in order to have 80% power to detect a difference of 5% or greater?

```
# To detect a 5% difference with SD = 8%, 80% power, alpha = 0.5
# Install 'pwr' package if not already installed
if (!requireNamespace("pwr", quietly = TRUE)) {
  install.packages("pwr")
}
library(pwr)
pwr.t.test(d = 5 / 8, power = 0.80, sig.level = 0.05, type = "two.sample", alternative = "two.sid
```

```
     Two-sample t test power calculation

              n = 41.16893
              d = 0.625
      sig.level = 0.05
```

```
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

Although part c for Q2 was not required, determing the appropriate sample size per group to detect a 5% difference in CD4+CD127+ expression with 80% power and a standard deviation of 8% would involve a two-sample power calculation. This ensures future studies are adequately powered to detect meaningful differences without being under- or over-resourced.

Part 2 text book problems from Chapter 11

Q3 Problem 5:

A crossover study was conducted to investigate whether oat bran cereal helps to lower cholesterol levels in hypercholesterolemic males. Fourteen such individuals were randomly placed on a diet that included either oat bran or corn flakes; after two weeks, their low-density lipoprotein (LDL) cholesterol levels were recorded. Each man was then switched to the alternative diet. After a second two-week period, the LDL cholesterol level of each individual was again recorded. The data from this study are shown below:

```
#| echo: false
# Load required library
if (!requireNamespace("DT", quietly = TRUE)) {
  install.packages("DT")
}
library(DT)

# Read the data from CSV
ldl_data <- read.csv("LDL_crossover_study.csv")

# Display the data as an interactive DT table
datatable(ldl_data,
          caption = "LDL cholesterol levels (mmol/L) for each subject after corn flakes and oat br
          options = list(pageLength = 14,
                         dom = 't'))
```

	Corn_Flakes	Oat_Bran	Difference
1	4.61	3.84	0.7700000000000005
2	6.42	5.57	0.8499999999999996
3	5.4	5.85	-0.4499999999999993
4	4.54	4.8	-0.2599999999999998

	Corn_Flakes	Oat_Bran	Difference
5	3.98	3.68	0.2999999999999998
6	3.82	2.96	0.8599999999999999
7	5.01	4.41	0.5999999999999996
8	4.34	3.72	0.6199999999999997
9	3.8	3.49	0.3099999999999996
10	4.56	3.84	0.7199999999999998
11	5.35	5.26	0.08999999999999986
12	3.89	3.73	0.1600000000000001
13	2.25	1.84	0.4099999999999999
14	4.24	4.14	0.1000000000000005

LDL cholesterol levels (mmol/L) for each subject after corn flakes and oat bran diets

a) Are the two samples paired or independent?

The two samples are paired, because each subject consumed both oat bran and corn flakes in a crossover study design. LDL levels were measured for the same individuals under both conditions, so the observations are not independent - they are matched within subjects.

b) What are the appropriate null and alternative hypotheses for a two-sided test

Let μ_D represent the **mean difference in LDL cholesterol** (Corn Flakes − Oat Bran) across all subjects.

- **Null Hypothesis (H_0):** $\mu_D = 0$
  There is no difference in LDL cholesterol between the two diets.
- **Alternative Hypothesis (H_1):** $\mu_D \neq 0$
  There is a difference in LDL cholesterol between the two diets.

c) Conduct the test at the 0.05 level of significance. What is the p-value?

We compute the difference in LDL cholesterol for each subject:

$$\text{Difference}_i = \text{LDL}_{\text{Corn Flakes},i} - \text{LDL}_{\text{Oat Bran},i}$$

Step 1: Differences

The differences (rounded to 2 significant figures for easy viewing) are:

```
0.77, 0.85, -0.45, -0.26, 0.30, 0.86, 0.60, 0.62, 0.31, 0.72, 0.09, 0.16, 0.41, 0.10
```

> *Note: Full-precision values were used for all subsequent calculations.*

Step 2: Mean of the Differences

$$\bar{d} = \frac{1}{n} \sum_{i=1}^{n} d_i = 0.362857 \text{ mmol/L}$$

Step 3: Standard Deviation of the Differences

$$s_d = 0.405964 \text{ mmol/L}$$

Step 4: Standard Error (SE)

$$SE = \frac{s_d}{\sqrt{n}} = \frac{0.405964}{\sqrt{14}} = \frac{0.405964}{3.741657} = 0.108498$$

Step 5: t-Statistic

$$t = \frac{\bar{d}}{SE} = \frac{0.362857}{0.108498} = 3.344$$

Step 6: 95% Confidence Interval

We use the critical value for a two-sided 95% confidence interval with 13 degrees of freedom (n -1 degrees, n = 14):

$$t^* = 2.1604$$

Margin of error:

$$ME = t^* \cdot SE = 2.1604 \cdot 0.108498 = 0.234397$$

Confidence interval:

$$\bar{d} \pm ME = 0.362857 \pm 0.234397$$

$$95\% \text{ CI} = [0.128461,\ 0.597254] \text{ mmol/L}$$

Step 7: Estimating the p-Value by Hand

We can assess statistical significance in two ways: using the **confidence interval** or by comparing our **t-statistic to a t-distribution table**.

Method 1: Use the Confidence Interval

The 95% confidence interval was:

$$[0.128461,\ 0.597254]$$

Since this interval does **not include 0**, it implies that the **two-sided p-value is less than 0.05**, and we reject the null hypothesis.

Method 2: Use a t-table to estimate the p-value

Given: $t = 3.3444$ and $df = 13$

From the t-table for 13 degrees of freedom:

- $t = 3.012$ corresponds to two-tailed $p = 0.01$
- $t = 3.852$ corresponds to two-tailed $p = 0.002$

Since $3.012 < 3.3444 < 3.852$, we estimate:

p is between 0.01 and 0.002

The **exact p-value**, calculated using R, is:

$p = 0.005278$

This falls within our range, confirming our t-table estimate but also illustrating the limits of printed tables — they only provide rough intervals, not precise probabilities.

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

T-table used to estimate p-value for t = 3.3444, df = 13 (SJSU 2007).

```
# R-code to check work:
# Perform paired t-test: Corn Flakes vs. Oat Bran
t.test(ldl_data$Corn_Flakes, ldl_data$Oat_Bran, paired = TRUE)
```

```
    Paired t-test

data:  ldl_data$Corn_Flakes and ldl_data$Oat_Bran
t = 3.3444, df = 13, p-value = 0.005278
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.1284606 0.5972537
sample estimates:
mean difference
      0.3628571
```

With R, the P-value is 0.005278.

d) What do you conclude?

Both methods confirm that $p < 0.05$, allowing us to reject H_0. The precise p-value from R, $p = 0.005278$, shows moderately strong evidence that oat bran lowers LDL cholesterol compared to corn flakes.

Q4 Problem 8:

A study was conducted to determine whether an expectant mother's cigarette smoking has any effect on the bone mineral content of her otherwise healthy child. A sample of 77 newborns whose mothers smoked during pregnancy has mean bone mineral content x-bar1 = 0.098 g/cm and standard deviation s1 = 0.026 g/cm; a sample of 161 infants whose mothers did not smoke has mean x-bar2 = 0.095 g/cm and standard deviation s2 = 0.025 g/cm. Assume that the underlying population variances are equal.

a) Are the two samples paired or independent?

The two samples are **independent**, not paired. Each measurement of bone mineral content comes from a different infant, and the infants are grouped based on whether their mothers **did** or **did not** smoke during pregnancy. There is no matching or repeated measurement across the groups, so a paired design is not appropriate.

b) State the null and alternative hypotheses of the two-sided test.

Let μ_1 be the mean bone mineral content in infants whose mothers **smoked**, and μ_2 be the mean in infants whose mothers **did not smoke**.

- **Null Hypothesis (H_0):** $\mu_1 = \mu_2$ (No difference in bone mineral content)
- **Alternative Hypothesis (H_A):** $\mu_1 \neq \mu_2$ (Difference in bone mineral content)

c) Conduct the test at the 0.05 level of significance. What do you conclude?

```
### Problem 8, Part c: Manual Two-Sample t-Test from Summary Stats
```

```
# Given values
x1_bar <- 0.098
s1 <- 0.026
n1 <- 77

x2_bar <- 0.095
s2 <- 0.025
n2 <- 161

# Pooled standard deviation
sp_squared <- ((n1 - 1) * s1^2 + (n2 - 1) * s2^2) / (n1 + n2 - 2)
sp <- sqrt(sp_squared)

# Standard error
```

```r
se <- sp * sqrt(1/n1 + 1/n2)

# t-statistic
t_stat <- (x1_bar - x2_bar) / se

# degrees of freedom
df <- n1 + n2 - 2

# two-sided p-value
p_value <- 2 * pt(-abs(t_stat), df)

# Output
list(t_statistic = t_stat, df = df, p_value = p_value)
```

```
$t_statistic
[1] 0.8549064

$df
[1] 236

$p_value
[1] 0.39347
```

> *Note:* We computed the t-statistic manually using the pooled standard deviation formula in R, as raw data were not available. This two-sample t-test assumes **equal variances**, as specified in the problem.

Problem 8, Part c: Conclusion

There was no statistically significant difference in bone mineral content between infants whose mothers smoked and those whose mothers did not (two-sample t-test, equal variances assumed, $t = 0.8549$, $df = 236$, $p = 0.3935$). We fail to reject the null hypothesis at the $\alpha = 0.05$ level.

Sources:

Pagano, M., & Gauvreau, K. (2018). Principles of biostatistics / Marcello Pagano, Kimberlee Gauvreau. Duxbury Press.

San Jose State University. (2007). T table. t-table. https://www.sjsu.edu/faculty/gerstman/StatPrimer/t-table.pdf

Thank you for reviewing and making it to the end of my work, in this assignment's signature end off I'd like to ask for feedback on what type of showing work method is preferred in this document. Does the mathmatical proof format used in problem 5 seem more professional or is the coding format in problem 8's calculation more legible. Let me know preference in the assignment comments. Best, Dylan